

GraphRAG Knowledge Workspace

For the AI developer implementing the redesign of `himanshusin/graph_rag · app.py`. Stack stays Streamlit. Light mode only. Primary user: analysts.

Reference mockups: [1a](#) workspace (recommended), [1b](#) console, [1c](#) command-first, [1d–1g](#) secondary screens.

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1 · Purpose & principles

The current UI is a dark, badge-heavy dashboard. Analysts use it to ask a question, read the answer, and check where each claim came from. The redesign optimizes for that loop and removes everything that competes with it.

Evidence is adjacent	Citations are never behind an expander. They sit beside (1a), beneath (1b) or inline under (1c) the prose. Every [n] in the answer is clickable and highlights its source.
One accent	Blue means “citation / current mode / primary”. Green means “verified / passed”. Amber means “partially grounded”. Nothing else is colored. No gradients, no glows, no emoji.
Hairlines over cards	Structure comes from 1px borders and whitespace, not from stacked shadowed cards. Shadows only on the composer and floating menus.
Numbers are mono	IDs, hashes, counts, weights, timings, page numbers and table cells use the mono face so analysts can scan and copy them.

Plain names	Drop “Enterprise”, “Executive”, “Strategic” prefixes. The five modes are Global, Local, DRIFT, Vector, Compare. See §7.
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2 · Tokens

Declare once as CSS custom properties in the injected stylesheet (§8) and reference everywhere.

Color

 --canvas	#F7F7F5	 --surface	#FFFFFF
 --surface-2	#F1F1EE	 --ink	#111110
 --ink-2	#55554F	 --ink-3	#8A8A83
 --line	#E8E8E4	 --line-strong	#D6D6D0
 --accent	#3056D3	 --accent-bg	#EEF2FD
 --ok	#1F8A5B · bg #E8F5EE	 --warn	#B7791F · bg #FBF3E4
 --danger	#B03A2E	 --highlight	#FFF0B3 (search match)

Community colors for the concept map and catalog, in order: accent, ok, warn, ink-3, danger, then #6B4FBB #2A8FA8 #C25E9A. Never more than 8 hues; group the tail as “Other”.

Type

--font-sans	Geist, then system-ui. Weights 400 / 500 / 600 only. Replace Plus Jakarta Sans.
--font-mono	Geist Mono, then ui-monospace. Weights 400 / 500. Replace JetBrains Mono.
Page title	26px / 1.2 / 500 / letter-spacing –.025em (only on 1c empty state)
Question	15px / 1.4 / 500
Answer body	14px / 1.6 / 400, bold lead-ins at 600, max line length 72ch
UI / table	12.5–13px / 1.45 / 400; labels 500
Meta	11–11.5px, ink-3. Section eyebrows 10.5px mono uppercase, letter-spacing .06em
Stat	20px mono 500, letter-spacing –.02em (1b strip)

Space, radius, elevation

Spacing	4-pt scale: 4 · 6 · 8 · 10 · 12 · 16 · 20 · 24 · 28 · 40. Page gutter 24px; thread gutter 40px.
Radius	4 badges · 6 nav/inputs · 7 buttons · 8 evidence cards · 10 panels/composer · 12 hero composer (1c) · 999 pills
Shadow	0 1px 2px rgba(0,0,0,.04) on composer; 0 8px 24px -12px rgba(0,0,0,.12) on hero composer and menus. Nothing else.
Motion	120ms ease-out for hover/border-color; 200ms for panel open. No spring, no bounce, no skeleton shimmer faster than 1.2s.

3 · Layout system

Recommended direction is [1a](#). Three fixed columns at ≥1200px, collapsing in order rail → sidebar.

Sidebar 212px	Brand + version, primary nav (Search, Vault, Concept map, Catalog, Governance), Reasoning mode list, Temperature and Citations sliders, QA status pinned to bottom. Surface white, right hairline.
Main flex	48px top bar (breadcrumb left, source filter pills right), scrollable thread constrained to 720px centered, composer pinned at bottom inside the same 720px column. Canvas background.
Rail 300px	Evidence for the <i>selected</i> assistant message (defaults to latest). Header “Evidence · n” with “Show context” toggle that reveals the raw context string sent to the LLM.
< 1200px	Rail becomes an inline expander under each answer (same card markup). < 900px: sidebar collapses to icons; mode moves into the composer selector.

Alternate directions: [1b](#) replaces sidebar with top tabs and a 5-cell stats strip, and makes the two-column Compare view the default answer layout; [1c](#) removes chrome entirely, with a 680px column, mode chips inside the composer, and evidence expanding inline beneath the sentence that cites it.

4 · Components

Nav item

32px tall, 8px horizontal padding, 14px stroke icon at 1.5px, 9px gap. Rest: ink-2. Hover: surface-2 background. Active: surface-2 + ink + weight 500. Optional trailing mono count in ink-3.

Mode selector

Radio semantics; one of Global · Local · DRIFT · Vector · Compare. Sidebar variant: 6px dot (filled accent when active, line-strong ring otherwise). Composer variant: 1px bordered chip with dot + label + chevron opening a menu that shows each mode's one-line description ("Synthesizes community reports across the whole corpus"). Segmented variant (1b): 2px inset track on canvas, active segment white with the 1px shadow.

Source filter pill

999 radius, 4×10 padding, 11.5px. Selected: ink background, white text. Unselected: 1px line-strong border, ink-2. Options: All sources, Documents, Graph, Tables. Filters which citation kinds are eligible; disabled kinds are struck through in the rail.

Message · user

24px circle initials avatar (surface-2 bg, ink-2 text) + question at 15px/500. No bubble.

Message · assistant

24px square "G" mark. Meta line above the body: *mode* in accent · n sources · elapsed (mono). Body is rendered markdown with [n] converted to <sup> links (10px mono, accent). Footer actions at 12px ink-3: Copy · Compare with vector search · Rerun · model name right-aligned.

Citation badge

Number in 10.5px mono 500, accent on accent-bg, 1×6 padding, radius 4. In 1c the inline variant is a pill "n · Kind"; kind ∈ Concept, Table, Passage, Domain, Relationship.

Evidence card

1px line border, radius 8, 10×12 padding. Row 1: badge, title (12.5px/500, "Kind · Name"), trailing mono meta (rank, page, chunk). Row 2 by kind — Domain/Concept: 12px ink-2 excerpt, 3-line clamp. Passage: excerpt in 12px mono on canvas background, radius 6. Table: mini table (mono 11px, header row on canvas, hairline rows, max 4 rows + "n more") and file name. Hover: border line-strong. Active (its [n] hovered or clicked): border #C9D3F5, background #FBFCFF. Click opens the source: Vault inspector for documents, Concept map for nodes, Catalog for domains.

Composer

White, 1px line-strong, radius 10, small shadow. Textarea auto-grows to 6 lines. Bottom row: mode chip, Attach (opens Vault ingest), send button 28px ink square with up-arrow. Enter sends, Shift+Enter newline. Helper line beneath, 11px ink-3, centered.

Grounding tag

Mono 10.5px pill: "grounded 96%". ≥90% ok colors, 70–89% warn, <70% danger. Computed as share of answer sentences that carry at least one citation. Shown on Compare columns and in the message meta when Compare is off.

Data table

White panel, 1px line, radius 10, overflow hidden. Header 11px ink-3 on #FAFAF8. Rows 12.5px with hairline separators, 10–11px vertical padding. Numeric columns mono and right-aligned. Search matches highlighted with --highlight. Footer: "Showing x of n · filtered by ...", page size, pagination. Replaces `st.dataframe` where the row count is small; keep `st.dataframe` for >500 rows.

Status dot

6px circle + label. ok = Indexed / QA passed; accent = in progress (paired with a 2px progress bar); warn = needs re-index; danger = failed.

Buttons

Primary: ink bg, white, 12–12.5px 500, 6×12 padding, radius 7. Secondary: white, 1px line-strong. Ghost/text: ink-3, hover ink. Danger is text-only in danger color; destructive actions confirm inline ("Delete doc_f0fc...? Delete · Cancel").

Slider

Label left, mono value right, 2px track (line) with ink fill, 10px white thumb with 1.5px ink ring.
Temperature 0–1 step .05, Citations 2–8.

5 · Screens & requirements

Search [1a](#) [1c](#)

- Maps to `tab_search_chat`. Keep `st.session_state.messages` schema; add `elapsed_s`, `model`, `grounding` per assistant message.
- Mode is chosen in sidebar or composer; both write the same state key. The four suggested prompts live in the empty thread only (1c list) and disappear after the first message.
- Each `[n]` in the answer resolves to `citations[n-1]`. Unresolvable numbers render plain, never as links.
- The rail shows evidence for the last assistant message; clicking an older message selects it (message gets a 2px left accent line while selected).
- “Compare with vector search” on any Global/Local/DRIFT answer re-runs the same query in Vector mode and renders the two-column Compare block below.
- Streaming: render tokens as they arrive; citations resolve when the stream ends. Show elapsed time live in the meta line.

Compare [1b](#)

- Two equal columns, each a panel: header (color square accent for graph / ink-3 for vector, title, source counts, grounding tag), body, footnote list of citations.
- Both runs execute concurrently; show each column’s skeleton independently.
- “Synced scroll” toggles linked scrolling; “Diff claims” (v2, optional) highlights sentences present in one column but not the other.
- Vector column must state explicitly when a sub-question is not covered by retrieved passages (ink-3 sentence), rather than omitting it.

Vault [1d](#)

- Maps to `tab_vault_ui` and `core/vault.py`. "Add document" opens a sheet with the uploader, "Re-index graph now" toggle (default on) and chunk-limit slider; replaces the expander.
- Registry columns: Document (title + mono id), Format, Size (human units), Pages, Tables, Uploaded (relative for <24h), Status. Sort by Uploaded desc. Selected row uses the active tint.
- Ingest progress is a row in the registry (accent dot, "Extracting k / n", 2px bar), driven by `st.status` steps: Persist → Extract text → Extract tables → Build graph → Sync Chroma.
- Inspector below the registry: metadata grid (SHA-256 full on hover/copy, storage path, characters, chunks in graph with "re-index" action, origin) and Extracted tables list (page, title, rows×cols) opening a full-width table view with row facts.
- Delete confirms inline and removes vault file, catalog entry, and Chroma docs; then clears caches.

Concept map [1e](#)

- Maps to `tab_concept_map`. Keep the PyVis embed; regenerate `interactive_graph.html` with a white background, the community palette from §2, edge color line-strong, node border 1px, physics on, and the Geist font for labels.
- Sidebar lists communities with color chip and node count; clicking filters the network (pass a query param or re-render the HTML with hidden nodes).
- Top bar: node count, edge count, algorithm; "Find node" search; size-by toggle Degree / Rank.
- Selected-node rail: title, type + community, three mono stats (degree, community, passages), description, relationships list with type and weight, actions "Ask about X" (prefills composer in Local mode) and "Open in catalog". Selection is passed from the iframe via `postMessage` or a query param; fall back to a node select box if not feasible.

Catalog [1f](#)

- Maps to `tab_catalog_ui`. Sub-tabs move into the top bar with mono counts: Concepts, Relationships, Nodes, Domain briefs.
- One search box filters the active tab across all text columns; matches are highlighted. Result count appears inside the box.
- Concepts columns: Title, Type (mono), Description (single line, ellipsis, full on hover), Degree (right), Community (chip + id). Relationships: Source → Target, Type, Weight (mono, /10), Description. Nodes: Title, Community, Degree. Domain briefs: rank-sorted list; clicking opens the full report in a right rail rather than an expander.
- Export CSV of the current filtered view.

Governance [1g](#)

- Maps to `tab_governance` and `core/change_manager.py`. Replace `st.json` with a check list: ✓/✗, check name, mono detail (rows, duration, path). Failed checks first, with the error text in a mono block beneath.
- Header shows last-run time; primary action "Re-run QA & sync" runs `run_full_sync()` with a step status, then reruns.
- Changelog parsed from `CHANGELOG.md` into a timeline: version (mono), date, "current" tag, commit rows (7-char sha + message, prefixes like `feat:` stripped), collapsed after 4 with "n more commits".

6 · States

Empty thread	Title "Ask the knowledge base.", composer, numbered list of the four suggested prompts with their default mode (1c). No welcome paragraph, no emoji.
Empty vault	Single centered line "No documents yet." + primary "Add document". Search composer is disabled with helper "Add a document to start".
Loading answer	Meta line shows mode + live timer; body shows three 12px-tall bars on surface-2 (widths 92 / 80 / 60%), rail shows two card skeletons. Replace the spinner text.
No results	Vector mode with zero passages above threshold: answer says so in one sentence and offers "Try Global synthesis".
Error	Inline in the assistant slot: danger dot, "Couldn't complete this search.", mono error summary, "Retry". Never <code>st.error</code> banners at page top.
Missing artifacts	Parquet or Chroma absent: Search shows the empty-vault state; Governance lists the failing checks. Concept map shows "No graph yet — add a document."

7 · Copy & naming

Sentence case everywhere. No exclamation marks. No emoji or icon fonts in copy. Rename as follows:

Current	New
Enterprise Knowledge Assistant	Knowledge

 Strategic Executive Synthesis	Global synthesis
 Targeted Concept & Dependency Lookup	Local lookup
 Deep-Dive Cross-Functional Analysis	DRIFT deep-dive
 Standard Document Passage Search	Vector passages
 Comparative Audit: Graph vs Document Search	Compare graph vs vector
Enterprise Document Vault / Retention Vault	Vault
Knowledge Concept Map	Concept map
Enterprise Data Catalog	Catalog
System Governance & Changelog	Governance
Verified Sources & Citations (n sources)	Evidence · n
Indexed Concepts / Verified Links / Retained Docs / Evidence Passages	Concepts / Relationships / Documents / Passages
● 98% VERIFIED GROUNDING (static)	grounded 96% (computed per answer; remove from header)
Response Precision (Temperature)	Temperature
Search enterprise knowledge or ask a strategic question...	Ask the knowledge base...

8 · Streamlit implementation

Theme. Add `.streamlit/config.toml`: `base="light" sc-camel-primary-color="#3056D3" sc-camel-background-color="#F7F7F5" sc-camel-secondary-background-color="#FFFFFF" sc-camel-text-color="#111110" font="sans serif"`. Keep `layout="wide"`, sidebar expanded.

Stylesheet. Move the CSS out of `app.py` into `ui/theme.css`, read once and inject with `st.markdown(<style>..., unsafe_allow_html=True)`. Import Geist and Geist Mono from Google Fonts. Define the §2 tokens as `:root` variables. Override Streamlit chrome: hide header [`data-testid="stHeader"`] and the footer, set `.block-container{padding:0 24px;max-width:none}`, restyle [`data-testid="stSidebar"`] to 212px white with right hairline, [`data-testid="stChatInput"`] to the composer spec, [`data-testid="stChatMessage"`] to no background and 24px avatars, `.stTabs` [`data-baseweb="tab"`] to the 2px-underline tabs, `.stSlider` to the 2px track.

Navigation. Use `st.navigation` with pages (Search, Vault, Concept map, Catalog, Governance) instead of `st.tabs` so each screen has its own URL and the sidebar nav is native. Move shared loaders into `ui/data.py`. The mode list, sliders and QA status render in `st.sidebar` beneath the nav via a custom-styled `st.radio` (hide the native radio circle, draw the 6px dot with CSS).

Three columns. `st.columns([1, 0.42], gap="large")` for thread + rail; the thread column wraps content in a `max-width:720px;margin:auto` div. The rail is an `st.container(height=...)` with the evidence cards rendered as HTML from a Jinja/f-string helper `ui/components.py: evidence_card(c)`. Under 1200px use a media query to stack the rail below the thread.

Citations. Post-process the LLM answer with a regex `\[(\d+)\]` → `<sup><a href="#cite-n" class="cite">[n]</a></sup>` only when `n ≤ len(citations)`; render with `unsafe_allow_html=True` after markdown conversion (use the markdown package with bleach sanitizing). Evidence cards carry `id="cite-n"`; hover-highlight is CSS `:target` plus a tiny `components.html` script for hover if wanted. Compute `grounding = cited sentences / total sentences`.

Streaming & timing. Use `st.write_stream(chain.stream(...))`; record `time.perf_counter()` around it and store elapsed on the message. Compare mode runs the two chains in a `ThreadPoolExecutor` and streams each into its own column placeholder.

Tables. Small registries (vault, QA checks, changelog, catalog ≤ 500 filtered rows) render as HTML tables from `ui/components.py` to hit the spec exactly; larger sets fall back to `st.dataframe` with `column_config` and the theme's font. Row selection uses `on_select="rerun"` where `st.dataframe` is used, or a hidden `st.selectbox` synced by query param otherwise.

Concept map. Regenerate the PyVis HTML in `core/pipeline.py` with `bgcolor="#FFFFFF"` `font_color="#111110"`, node colors from the §2 palette by community, edges `#D6D6D0`, and inject a `selectNode` handler that sets `window.parent.location.search=?node=ID`; read `st.query_params["node"]` to fill the rail.

Known limits. Streamlit cannot pin the composer to the viewport bottom inside a column; accept `st.chat_input`'s native fixed position and restyle it to 720px centered. Segmented control (1b) is `st.segmented_control`. Keyboard ⌘K focus needs a small `components.html` listener. If custom HTML tables become a maintenance burden, keep `st.dataframe` and apply only the typography tokens.

File plan. `app.py` (entry + navigation) · `pages/search.py` `vault.py` `concept_map.py` `catalog.py` `governance.py` · `ui/theme.css` `ui/components.py` `ui/data.py` · `.streamlit/config.toml`. Core engines unchanged.

9 - Accessibility

- Body text ink on canvas/surface $\geq 12:1$; ink-2 $\geq 7:1$; ink-3 is for $\geq 11\text{px}$ meta only (4.6:1 on white). Accent on white is 5.9:1; accent on accent-bg 5.4:1.
- Status is never color-only: every dot has a text label; grounding tags include the number.
- Citation links have `aria-label="Source n: title"`; evidence cards are `<article>` with headings.
- All controls reachable by keyboard; focus ring 2px accent at 2px offset. Hit targets $\geq 32\text{px}$ on desktop.
- Respect `prefers-reduced-motion`: disable skeleton animation and physics autoplay in the map.

10 - Acceptance checklist

1. No emoji, gradients, glows or uppercase-tracked stat labels remain anywhere in the UI.
2. Fonts render as Geist / Geist Mono; only weights 400, 500, 600 are used.
3. Every `[n]` in an answer links to a visible evidence card; hovering highlights it; clicking opens the source screen.
4. Mode can be changed from sidebar and composer and both stay in sync.
5. Compare renders two columns concurrently with grounding tags and footnotes.
6. Vault registry shows ingest progress as a row and the inspector below; delete confirms inline.
7. Concept map is white with the §2 palette; selecting a node fills the rail; "Ask about X" prefills the composer in Local mode.
8. Catalog search highlights matches and reports "Showing x of n".
9. Governance shows checks as a list with failures first; changelog is a timeline.
10. Empty, loading, no-result and error states match §6; no top-of-page `st.error` banners.
11. Layout holds at 1440, 1200 and 1024px widths without horizontal scroll.